

Optimum Sound Attenuation in Flow Ducts Based on the “Exact” Cremer Impedance

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Summary

The Cremer impedance (Acustica 3, 1953) [1] is the locally reacting boundary condition that maximizes the attenuation of a certain mode in a uniform wave guide taken as the lowest order mode or “plane” wave. This paper presents the analysis of the “exact” Cremer impedance model, i.e., the high frequency asymptotic results proposed by Tester for uniform mean flow (JSV 28(2), 1973) [2] are extended to lower frequencies. It is shown that significantly larger attenuation per unit length can be obtained using the exact instead of the asymptotic solution. However, for sufficiently low frequencies the “exact” Cremer solution and optimum attenuation is requiring a wall impedance with a negative real part, i.e. an active boundary. In addition, the effect of a finite length on the resulting attenuation is studied using a finite element method for solving the convected wave equation. Finally, it is demonstrated how a silencer can be built that realize the optimum Cremer impedance at a given frequency by using a micro-perforated panel and locally reacting cavities. The performance of the optimized silencer is determined experimentally and the results are compared to the prediction of the finite element model.

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1. Introduction

Dissipative silencers are classically based on fibrous materials. However, applying such materials in flow-ducts can have many drawbacks, e.g., emitting airborne fibers or unpredictable deterioration of acoustic performance. These and other drawbacks of fibrous materials is an issue in many applications and environments, e.g., air-conditioning systems of buildings and vehicles and the gas exchange system of the internal combustion (IC) engines etc. In addition, the European Commission has stipulated strict regulations for fibrous materials to be used in the gas exchange system of the motor vehicles [3].

One of the alternatives, with potential of substituting fibrous materials for dissipative noise control, is the micro-perforated panel (MPP) [4, 5, 6]. In general, the dissipation of the MPP is caused by the viscous losses of the relatively small size apertures and flow interaction effects. In addition, non-linear losses can arise in case of high particle velocity inside the aperture. However, in order to effectively utilize the MPP losses, design has to be fine-tuned to specific acoustic field making an optimization strategy important.

The authors recently proposed a dissipative compact MPP silencer for the control of IC engine compressor noise [7]. The concept was based on the findings from earlier investigation on the use of the MPP for noise control in ducts [8, 9, 10]. The proposed silencer concept is based on an MPP tube with adjoining locally reacting air cavities (See Figure 1).

The optimization was suggested [7] to be carried out by matching the acoustic impedance of the locally reacting silencer wall to the Cremer impedance [1]. In order to adjust for the mean flow the asymptotic solution valid for high Helmholtz-numbers (“well above cut-on”), as suggested by Tester [2], was adopted in [7]. This approach is questionable in applications with appreciable mean flow, and where the low frequency range is important, e.g., IC-engine noise which is a motivation for this study.

In [11], Bravo *et al.* studied the same silencer concept as proposed in [7, 8, 9, 10], in the iterative optimization studies of MPP silencers in absence of mean flow, and in linear and non-linear regimes. They also compared the result obtained by their method to the Cremer optimal impedance model and concluded, that their optimization method of finite length silencer leads to the Cremer impedance in the plane wave frequency range, except in a very low frequency range where higher reflections are expected.

In the present work, the Cremer optimal impedance model is analyzed in presence of mean flow and without

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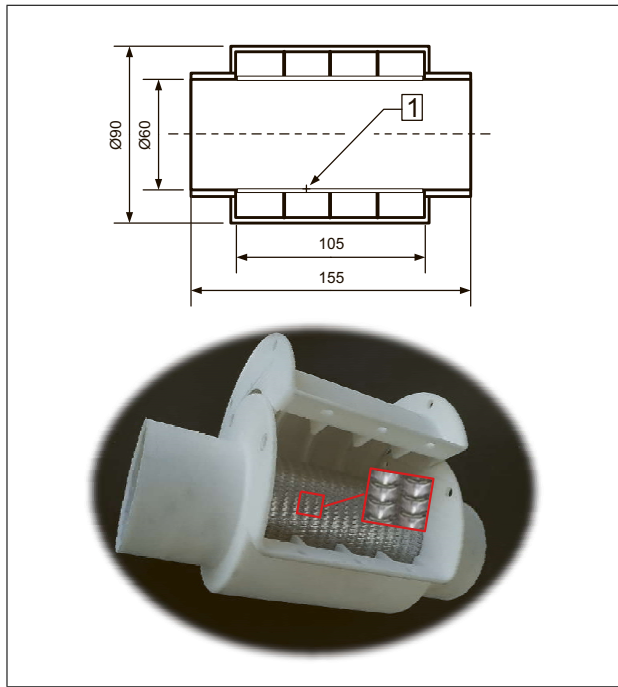


Figure 1. Sketch and a photo of the compact silencer prototype [7]. The prototype consists of an Acustimet™ MPP tube (“1”) and adjoining cylindrical cavities separated by side walls to create a locally reacting surface.

Testert’s “well above cut-on” assumption. Theoretical results are obtained under the assumption of an infinite wave guide, and complementary finite element analysis (FEA) are conducted for finite length duct cases.

2. The Cremer optimal impedance model

The specific problem considers the propagation and attenuation of sound in an infinite uniform two-dimensional wave guide, with a locally reacting wall impedance and a uniform (plug) flow, see Figure 2. The boundary condition is the classical Ingard–Myers condition [12, 13], i.e., the acoustic normal displacement and the acoustic pressure are continuous at the wall.

The propagation of sound in an axisymmetric wave guide can be described by superposition of complex acoustic modes of well-known form,

$$p_n = A_n e^{i(\omega t - k_{x_n} x)} J_0(k_{r_n} r), \quad (1)$$

where p_n is the acoustic pressure of the n th acoustic mode, in Pa. A_n is the amplitude of the n th acoustic mode, in Pa, i represents the imaginary unit, ω the angular frequency, in Hz, t is time in s. k_{x_n} is the longitudinal wave number of the n th mode, in 1/m, x is the longitudinal co-ordinate in m, k_{r_n} is the radial wave number of the n th mode in 1/m and r is the transverse co-ordinate in m.

Following Cremer and Tester [1, 2] it is assumed that the least attenuated acoustic mode in the problem is the 0th order or so called fundamental plane wave mode. Hence the mode number n is abandoned in the further expressions. In

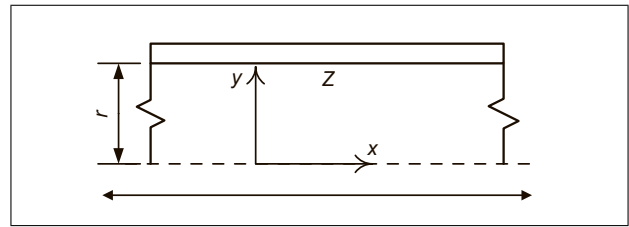


Figure 2. Sketch of the studied uniform 2D wave guide with a locally reacting wall, where Z represents the specific acoustic wall impedance, r the radius of the circular wave guide, x is the longitudinal co-ordinate axis and y (or r) the transverse or radial co-ordinate axis.

addition, the specific acoustic impedance of the air is used to normalize all the acoustic impedance expressions.

The damping of the mode is directly related to the axial decay rate of the propagating acoustic mode in the wave guide and is given by the imaginary part of the corresponding longitudinal wave number. The expression of the normalized longitudinal wave number in the wave guide is

$$\frac{k_x}{k} = \frac{-M_x \pm \sqrt{1 - (k_r/k)^2(1 - M_x^2)}}{1 - M_x^2}, \quad (2)$$

where k_x is the longitudinal wave number, 1/m, k_r the radial wave number, 1/m, k the wave number (ω/c), 1/m, c represents the speed of sound, m/s and M_x is the Mach number of the uniform mean flow.

By assuming no dissipation in the medium, i.e., k is real, the imaginary part of the longitudinal wave number k_x is controlled only by the radial wave number k_r as the only complex quantity in the longitudinal wave number expression (Equation 2). The existence of an optimal radial wave number, leading to the maximum axial decay rate of the propagating acoustic zeroth mode was first addressed by Cremer [1] for rectangular wave guides.

In 1973 Tester [14] further generalized the work by Cremer to any acoustic mode and included both annular and circular wave guides. In addition, it was shown that under optimum condition two sequential acoustic modes are merging in the complex k_r plane, thus making the optimal transverse wave numbers a branch point of the wave guide eigenvalue equation. Moreover, the effect of a uniform or “plug” flow was also included by Tester [2].

By following Tester’s work [2], the optimal wave numbers for a circular uniform wave guide, carrying a uniform mean flow, are the solutions of the corresponding branch point equation,

$$\frac{d}{d(k_r r)} \left[\left(1 - M_x \frac{k_x}{k} \right)^{-2} \frac{k_r r J'_0(k_r r)}{J_0(k_r r)} \right] = 0. \quad (3)$$

From the preceding branch point equation, it is clear that the optimal radial wave number is frequency and Mach number dependent. In order to significantly simplify the derivation of the optimal impedance, the high frequency range assumption of “well cut-on” modes was proposed

by Tester [2]. This assumption corresponds to

$$\frac{k_r}{k}(1 - M_x)^2 \rightarrow 0, \quad (4)$$

and thus

$$k_x = \frac{k}{1 \pm M_x}. \quad (5)$$

By applying the “well cut-on” frequency range assumption, i.e., Equation (5), to the branch point equation (Equation 3), a significantly simplified branch point equation is obtained:

$$\frac{d}{d(k_r r)} \left[(1 \pm M_x)^2 \frac{k_r r J'_0(k_r r)}{J'_0(k_r r)} \right] = 0. \quad (6)$$

The wave number resulting from Equation (6) is independent of frequency and Mach number i.e. the solution with this approximation is identical with the no flow case. Based on this, the optimal impedance for non-zero mean flow simply involves the zero mean flow impedance [2],

$$Z_{\text{opt}} = (0.88 - 0.38i) \frac{kr}{\pi}, \quad (7)$$

corrected by the term $(1 \pm M_x)^2$ as

$$Z_{\text{opt}}^A \big|_{M_x} = \frac{Z_{\text{opt}}}{(1 \pm M_x)^2}, \quad (8)$$

where Z_{opt} is the optimal impedance for a circular wave guide in absence of mean flow and Z_{opt}^A is the asymptotic optimal impedance for a circular wave guide.

In the present work, the high frequency assumption used by Tester [2], is eliminated, and the exact expression for the longitudinal wave number (Equation 2) is used instead in the branch point equation (Equation 3). This modification of the Tester high frequency asymptotic model is vital for obtaining accurate optimal radial wave numbers, especially in the low frequency range and under appreciable Mach number conditions, where the deviation from high frequency asymptotic model is expected to be the most significant.

The exact longitudinal wave number expression (Equation 2) contains a square root with two possible sign options. To choose the correct sign the approach suggested by Eversman [15] is used herein. The approach is based on the assumption of propagating acoustic modes with decaying amplitude in their direction of propagation, i.e., the imaginary part of the longitudinal wave number must be negative in case of spatial dependence of $-ik_x x$ and $\Re(k_x) > 0$ (see Equation 1). Therefore, if the principal square root function is evaluated in the upper half of the complex plane, i.e., with positive imaginary part, the minus sign in the Equation (2) has to be selected consistently throughout the analysis.

By defining the mean flow related term in the branch point Equation (3) as

$$f_1 = 1 - M_x \frac{k_x}{k}, \quad (9)$$

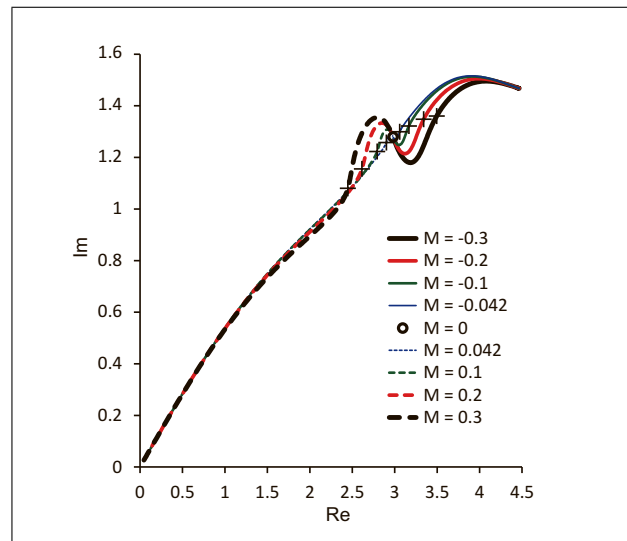


Figure 3. Theoretical optimum radial wave number trace (for maximum attenuation) for varying Helmholtz-number (kr) and different Mach-numbers (M) computed from Equation (10). The + denotes the “cut-on” Helmholtz-number (kr) of first higher order mode in hard walled duct.

and carrying out essential manipulations (See Appendix), the branch point equation takes the form

$$\frac{df_1}{d(k_r r)} \frac{2J_1(k_r r)}{f_1} - J_0(k_r r) - \frac{J_1^2(k_r r)}{J_0(k_r r)} = 0, \quad (10)$$

where

$$\frac{df_1}{d(k_r r)} = \frac{-M_x k_r r}{(kr)^2 \sqrt{1 - (k_r/k)^2 (1 - M_x^2)}}. \quad (11)$$

Computing the optimal radial wave numbers involves finding the roots to the non-linear transcendental equation (Equation 10). Here the “vpasolve” solver from Matlab® is used with initial values defined by the asymptotic (“high frequency”) known solution. In general, an unlimited number of solutions and their complex conjugates satisfying the Equation (10) can be found. These solutions correspond to increasing order of sequential acoustic mode pairs merging at a branch point in the plane. Nevertheless, only the first solution corresponding to the first radial mode pair, referred to by Tester [2] as “(0, 1)”, is considered here to maximize the longitudinal decay rate of the zeroth or “plane wave” mode. Once the optimal radial wave numbers are available, the attenuation level of time-averaged acoustic power per duct radius (r) can be computed as

$$TL_\infty' = 20 \log_{10} [e^{-\Im(k_x)r}]^{-1}. \quad (12)$$

The “exact” radial wave number traces on the complex $k_r r$ plane, computed for different Mach numbers by varying the Helmholtz number kr , are presented in Figure 3.

The high frequency asymptotic model by Tester is based on the constant radial wave number $k_r r$ value of (3.0 +

1.3i) [2]. A very close value of as high frequency asymptote or zero-mean flow value is obtained from the numerical solution here. This constant value can be observed as a point in the Figure 3. In contrast, a large variation of optimal radial wave number $k_r r$ can be observed in case of non-zero mean flow, i.e., a significant deviation of optimal wave guide wall impedance from the constant value based high frequency model (Equation 8) is expected. In addition, the existence of low frequency asymptotic values, $(0 + 0i)$ for positive Mach number and $(4.47 + 1.47i)$ for negative Mach number, are evident from the Figure 3. As a consequence of the existence of high and low frequency asymptotic values, a different mean flow Mach number changes only the path on the complex plane between these two points. This Mach number dependent difference appears to be the most significant near the high frequency asymptotic value and becomes negligible in low frequency range.

The TL spectra in Figure 4 are based on the imaginary part of the longitudinal wave number k_x , computed according to the Equation (2). As was mentioned previously, the imaginary part is controlled by the radial wave number k_r as the only complex quantity in the Equation (2). Therefore, a meaningful comparison between the exact and asymptotic (constant) radial wave number based models cannot be produced as the TL of the later becomes independent of flow direction ($\pm M_x$). Although similar behavior of the exact model can be observed in the asymptotic high frequency range (Figure 4), for $kr < 4$ a significant difference exists between the up- and downstream directions.

The corresponding optimal acoustic impedance of the locally reacting wall is computed by utilizing Ingard-Myers boundary condition and the previously obtained optimal radial wave number as

$$Z_{\text{opt}}|_{M_x} = \frac{i\pi(1 - M_x k_x/k)^2 J_0(k_r r) k_r}{k_r r J_1(k_r r)} \frac{kr}{\pi}. \quad (13)$$

The optimum Cremer impedance of the “exact” and asymptotic models [2] are compared in Figure 5. In general, a clear difference between the optimal impedance spectra of the asymptotic and the “exact” models can be observed. The difference between the models is predictably increasing with mean flow Mach number and towards lower Helmholtz numbers. From the perspective of a passive noise control, a rather undesirable feature of negative resistance can also be observed in Figure 5. The low frequency band of negative resistance is Mach number dependent, and will be wider, i.e. reaching to higher frequencies, in case of higher Mach number values. In this frequency band the wall of the wave guide has to be active in order to produce the maximum attenuation of the propagating wave. The transition values of Helmholtz numbers where the resistance changes from positive to negative, i.e., zero resistance, are also denoted in Figure 4 to show the low Helmholtz number limit of a passive wave guide wall.

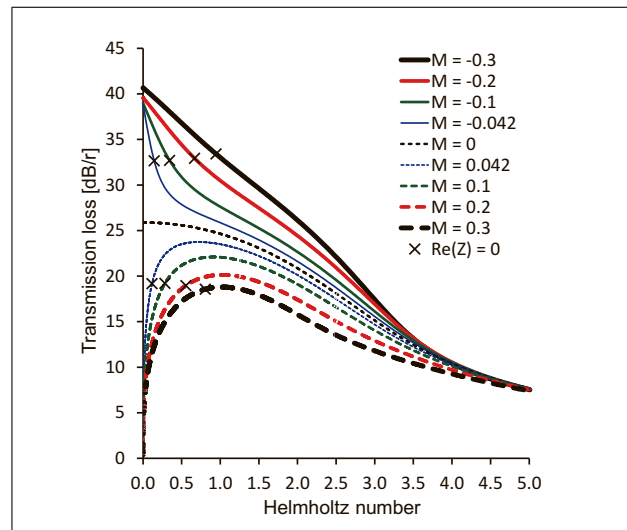


Figure 4. Theoretical maximum attenuation rate per duct radius (r) based on the “exact” radial wave number. The X denotes the Helmholtz-number where the real part of the Cremer impedance becomes negative (see Figure 5).

3. Finite element analysis

The theoretical modeling results in the previous section are valid for a uniform wave guide with infinite length, Figure 2. Here the effect of a finite length and reflection at the inlet/outlet as well as scattering to other modes will be investigated. Two different lengths l_r of finite duct are considered: $100r$ approximating the theoretical infinite duct case, and $5r$ where the reflective boundaries are closer.

The FEA computations are carried out by means of the commercial software COMSOL Multiphysics® [16]. The linearized acoustic field in a lossless medium with a mean flow (assumed to be potential) is determined by solving the equation

$$\begin{aligned} & -ik(k\Phi + M_x \cdot \nabla\Phi) \\ & + \nabla[\nabla\Phi - (ik\Phi + M_x \nabla\Phi)M_x] = 0, \end{aligned} \quad (14)$$

where Φ is the acoustic velocity potential, m^2/s .

The geometry of the studied wave guide (see Figure 2) is axisymmetric, hence an efficient truncated 2D modeling domain with axisymmetric boundary conditions is used. The resulting modeling domain with an example of the acoustic pressure distribution is presented in Figure 6.

The locally reacting wall impedance boundary condition is simulated at the outer radius of the modeling domain using the Ingard-Myers condition implemented in COMSOL Multiphysics® [16]. In addition, an 8 cm long section ($2.66r$) with a so called “sound hard” boundary condition are used before and after the locally reacting segment (See Figure 6) by setting the velocity in radial direction to zero. In longitudinal direction at both terminations (inlet/outlet) reflection free boundary conditions are imposed with an incident plane wave at the inlet side ($x = 0$).

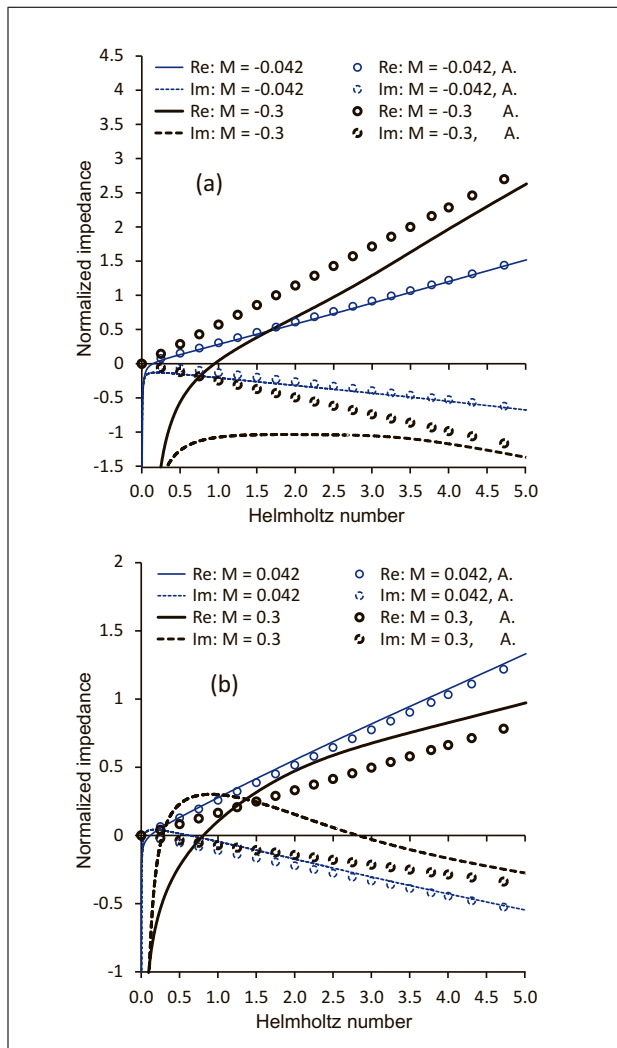


Figure 5. Theoretical optimum Cremer impedance for a circular wave guide wall for negative mean flow direction (See Figure 2) – a, and for positive direction – b. “A.” denotes the asymptotic model [2] (Equation 8).

The computation grid consists of regular 2D rectangular shaped elements. Although, satisfactory convergence is obtained by using as few as five elements per highest frequency wave length, the conservative approach of twenty five elements per wave length is used herein. Nevertheless, the model still remains inexpensive in terms of computational resources.

Once the velocity potential field is computed, time averaged acoustic power can be obtained from the incident (imposed) and transmitted (computed) velocity potential. The acoustic transmission loss (in the plane wave range) is given by the square of the velocity potential amplitude,

$$TL_{FEM} = 20 \log_{10} \frac{|\Phi_{in}|}{|\Phi_{out}|}, \quad (15)$$

where the subscripts in/out refer to the incident/transmitted wave amplitudes. The transmitted amplitude is computed as the surface average at the reflection free termination of the modeling domain. To compare with the previ-

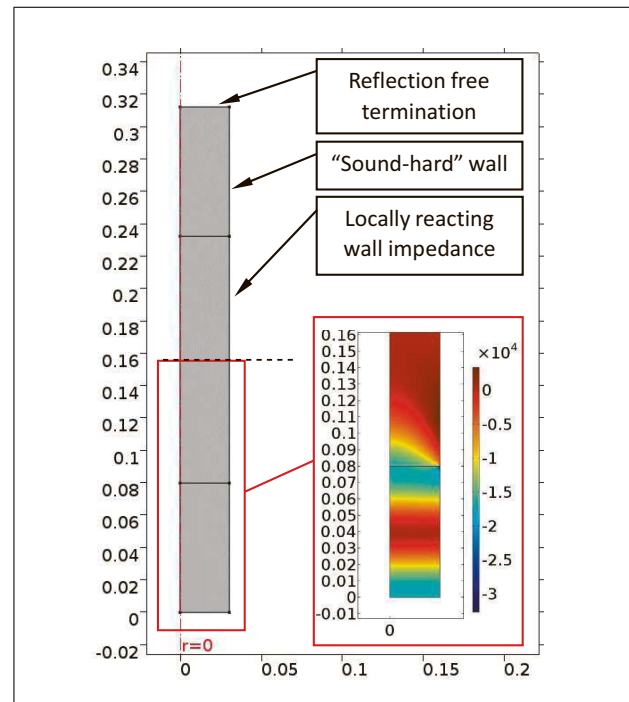


Figure 6. The modeling domain of the finite element analysis with indicated locations of boundary conditions and acoustic pressure distribution plotted on the first half of the modeling domain. Length scale is in meters and color legend is in Pascals.

ous results for an infinitely long wave guide (Equation 12) the transmission loss is reduced to a length of $1r$,

$$TL_{FEM}^r = \frac{r TL_{FEM}}{l_r}. \quad (16)$$

The transmission loss spectra of the asymptotic and the FEM models (with exact Cremer impedance) for different finite duct lengths reduced to unit length (r) are compared in Figure 7. In general, the TL performance appears to be remarkably improved over the asymptotic model. Expectedly, the most significant improvement occurs in the low frequency range, and is increasing with mean flow velocity. However, it should be recalled that in the low frequency range the requirement of an active wall, i.e. $\Re(Z)$, appears (Figure 5). The curves for the $100r$ case are virtually identical to the results plotted in Figure 4, i.e. the theoretical infinite wave guide case. The modest reduction obtained when using a realistic duct length of $5r$ demonstrates a high potential to use Cremer optimized duct linings to create efficient silencers in the plane wave range ($kr < 1.8$).

4. Realization of the Cremer impedance

In the present work the concept proposed by the authors in [7] is treated. This concept realizes the optimal (“Cremer”) wall impedance at a single frequency by the combination of a MPP and a locally reacting cavity (See Figure 1). The concept is based on two steps. First, the real part of the Cremer impedance at the chosen frequency is fixed to the

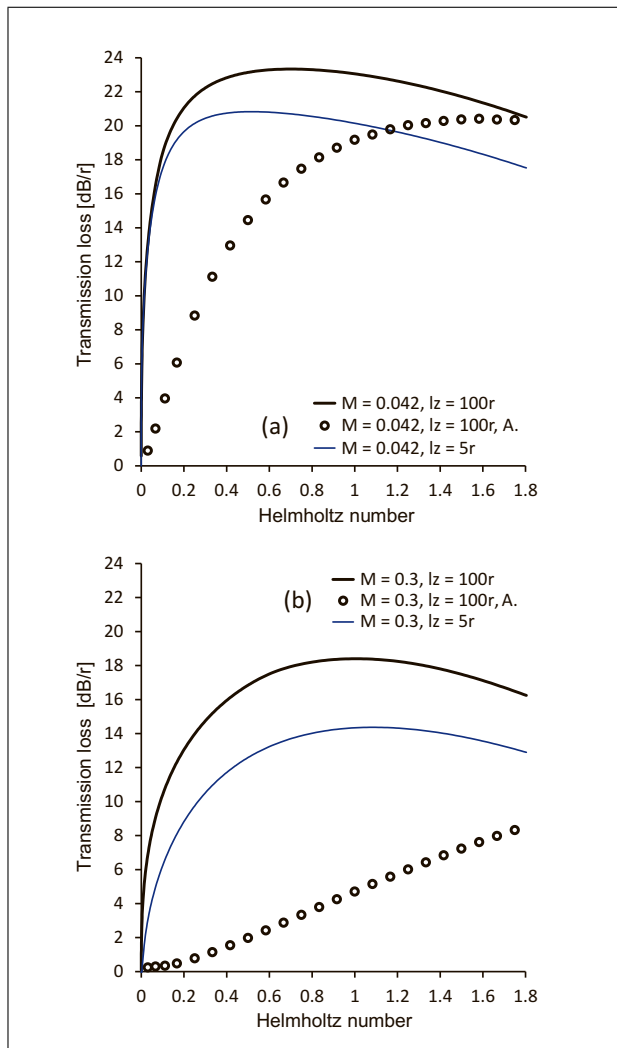


Figure 7. Comparison of asymptotic (A. – Equation 8) and FEM model (exact Cremer imp.) TL per duct radius (r) under two different mean flow conditions. Mach = 0.042 – a, and Mach = 0.3 – b.

correct value by selecting a suitable MPP. Secondly, the imaginary part is adjusted by choosing a suitable cavity depth.

4.1. Impedance model for the locally reacting wall

The impedance of the locally reacting wall is a combination of the MPP transfer-impedance and a locally reacting air cavity impedance,

$$Z_{\text{wall}} = Z_{\text{MPP}} + Z_{\text{cav}}, \quad (17)$$

where Z_{wall} is the locally reacting wall impedance, Z_{MPP} the MPP transfer impedance and Z_{cav} the locally reacting air cavity impedance.

For the MPP used in this work the transfer-impedance of a slit-holed plate is required [8]:

$$\Re(Z_{\text{MPP}}) = \Re \left\{ \frac{i\omega t}{\sigma c} \left[1 - \frac{\tanh(k_s \sqrt{i})}{k_s \sqrt{i}} \right]^{-1} \right\} + \frac{4R_s}{\sigma c} + \frac{|u_h|}{\sigma c} + \beta \frac{|M|}{\sigma}, \quad (18)$$

and

$$\Im(Z_{\text{MPP}}) = \Im \left\{ \frac{i\omega t}{\sigma c} \left[1 - \frac{\tanh(k_s \sqrt{i})}{k_s \sqrt{i}} \right]^{-1} \right\}, \quad (19)$$

where t represents the MPP thickness in m, σ is the perforation ratio, k_s is the shear wave number, R_s the surface resistance, Pa s/m, u_h is the peak particle velocity inside apertures, m/s, β the coefficient of grazing (tangential) flow effects and M the grazing flow Mach number.

The shear wave number in Equations (18) and (19), which relates the aperture dimensions with the acoustic boundary layer thickness, is defined as

$$k_s = d_s \sqrt{\frac{\omega}{4\nu}}, \quad (20)$$

where d_s is the slit width, m and ν the kinematic viscosity, m^2/s .

The MPP resistance in the Equation (18), which is a consequence of oscillating motion of the fluid on the surface of a perforated panel, can be obtained as

$$R_s = \sqrt{\frac{\omega\nu}{2}}. \quad (21)$$

Based on the experimental investigation on MPP slit-holed plates (Acustimet™) in [8], a new best fit for the flow effects, based on extrapolating the data to higher frequencies, will be used here. The frequency dependent coefficient β for the MPP used in the prototype (Figure 1) is

$$\beta(\omega) = -2.29 \cdot 10^{-10} \omega^2 + 1.02 \cdot 10^{-6} \omega + 0.07022$$

Similarly a best fit to the MPP reactance data in [17] for the MPP used gave

$$\Im(Z_{\text{MPP}}) \big|_M = \frac{\Im \left\{ \frac{i\omega t}{\sigma c} \left[1 - \frac{\tanh(k_s \sqrt{i})}{k_s \sqrt{i}} \right]^{-1} \right\}}{1 + (10|M|)^3}, \quad (23)$$

which is valid up to $M < 0.15$.

Finally, the impedance of the locally reacting (axisymmetric) cavity is computed according to following analytical model [10]:

$$Z_{\text{cav}} = \frac{i \left[H_0^{(1)}(kr) - \frac{H_1^{(1)}(kR)}{H_1^{(2)}(kR)} H_0^{(2)}(kr) \right]}{H_1^{(1)}(kr) - \frac{H_1^{(1)}(kR)}{H_1^{(2)}(kR)} H_1^{(2)}(kr)}, \quad (24)$$

where $H_m^{(n)}$ is the Hankel function of n th kind and m th order, R is the outer radius of the cavity, m.

Note, in order to ensure local reaction of the wall, the total cavity length must be divided into sections with intermediate distance between baffles less than half a wave length at the highest frequency of interest [10].

The locally reacting wall impedance Z_{wall} should now be tuned to match the Cremer optimal impedance. However, because of the difference in frequency variation this

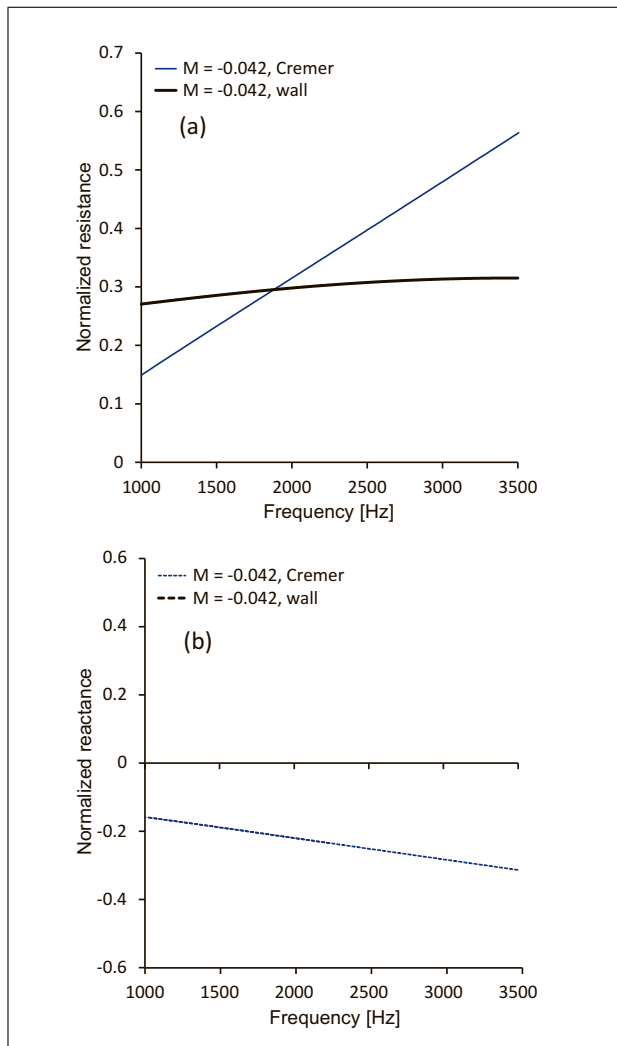


Figure 8. Comparison of wall impedance and “exact” Cremer impedance for the prototype (a – resistance and b – reactance).

Table I. Geometrical parameters of the silencer prototype studied. R : outer cavity radius; r : wave guide (duct) radius; t : MPP thickness; d_s : MPP slit width and σ : MPP perforation ratio.

R [mm]	r [mm]	t [mm]	d_s [mm]	σ %
45	30	1.2	0.24	6.5%

can only be achieved at one frequency referred to as target frequency of the optimization.

The MPP data and chamber geometrical parameters for the prototype silencer studied (see Figure 1) are listed in Table I and the resulting wall impedance is compared to the “exact” Cremer impedance in Figure 8.

As can be seen the prototype is not giving a perfect match to the Cremer impedance at a single frequency, the real parts are equal around 1900 Hz and the imaginary parts around 2500 Hz.

4.2. Experimental validation

The prototype silencer is modelled numerically using the FEA approach, described earlier, and the results are com-

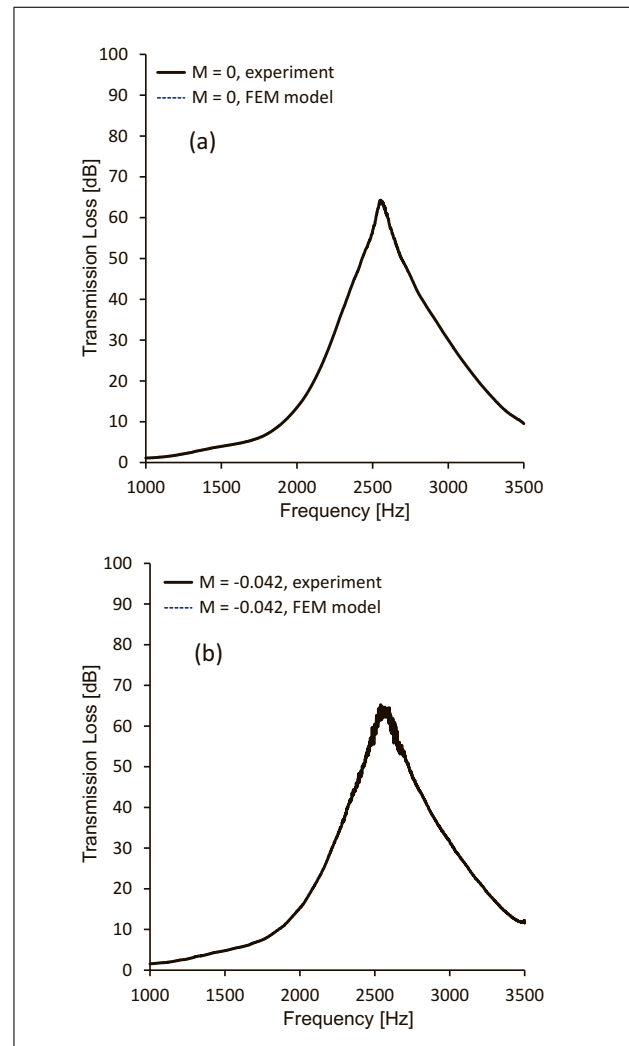


Figure 9. Experimental validation of the FEM model in absence of mean flow – a, and under -0.042 Mach number mean flow conditions – b.

pared with experimental data. First, the acoustic two-port [18] data are determined, then the transmission loss (TL) is computed and compared with the simulated TL. More details on the experiments can be found in [19].

To improve the FEA model also the three baffles between the four cavities in the outer chamber (see Figure 1) are simulated by adding short “sound hard” segments to the locally reacting wall impedance (Equation 17) boundary condition (see Figure 6).

Experimentally determined TL spectra of the silencer prototype are presented and compared to the FEA model predictions in Figure 9.

The silencer prototype provides a very high level of TL close to the matching frequency of the optimum reactance. Moreover, an appreciable level of TL is maintained across a wide frequency band. The previous observations are true in the absence of flow (Figure 9a) as well as in presence of a moderate mean flow (Figure 9b). But it can be noted that the FEA model is predicting an even higher TL peak value than the one actually measured. This over-prediction occurs because of a very high peak value (>60 dB), which is

difficult and unrealistic to achieve in experimental setups due to flanking transmission. Such phenomena is not included in the FEA model, and hence the over predicted TL peak value of $\sim 80\text{--}90\text{ dB}$ is not experimentally achieved. Except for a narrow frequency band in the vicinity of the peak TL value, the FEA model predicts TL spectra of the silencer prototype remarkably well.

The result also demonstrates the efficiency of the proposed silencer concept (see Figure 1) and shows that an exact match of the real part of the Cremer impedance is not critical. Furthermore, the design can be made more robust by tuning different parts of the silencer to different target frequencies, i.e., the use of several target frequencies where the Cremer impedance is matched. This can as is exemplified in [7] widen the attenuation band significantly.

5. Dissipation versus reflection in the optimal design

In a uniform wave guide no longitudinal reflection occur as the impedance boundary condition is continuous, i.e. the attenuation is caused by the dissipation of acoustic power at the wave guide wall. On the contrary, in a finite length flow-duct, like the compact silencer in Figure 1, longitudinal reflection does occur at the interfaces of wall impedance discontinuities, i.e. at the inlet and outlet. As a consequence the total transmission loss is caused by the reflection and dissipation of time averaged acoustic power. In order to evaluate if dissipation or reflection dominates the TL behavior, the transmission loss associated with dissipation can be determined as

$$TL_{\text{dis}} = 10 \log_{10} \left[\frac{1 - |R|^2 \frac{(1-M_x)^2}{(1+M_x)^2}}{|T|^2} \right], \quad (25)$$

where R is the reflection coefficient of complex acoustic pressure wave amplitude, T the transmission coefficient of complex acoustic pressure wave amplitude.

From this it follows that the TL that is caused only by the reflection is given by

$$TL_{\text{ref}} = -10 \log_{10} \left[1 - |R|^2 \frac{(1-M_x)^2}{(1+M_x)^2} \right]. \quad (26)$$

Note, that by adding Equations (25) and (26), the total TL, i.e., the classical definition of TL, is obtained.

In Figure 10 and 11, this analysis is applied to the cases studied in the section 2 and 3 (Figure 7b and 9b). It can be observed in Figure 10 that the simulated TL of the asymptotic and exact models is predominantly caused by dissipation (Figure 10b). Moreover, a significantly improved TL of the exact model is achieved via increased level of dissipation and not the reflection, regardless the lower level of wall resistance (Figure 5b). Although, the reflection is significantly increased in case of realistic duct length of $5r$ (Figure 10c and 10b), the level still remains negligible compared to the dissipation. In other words the Cremer

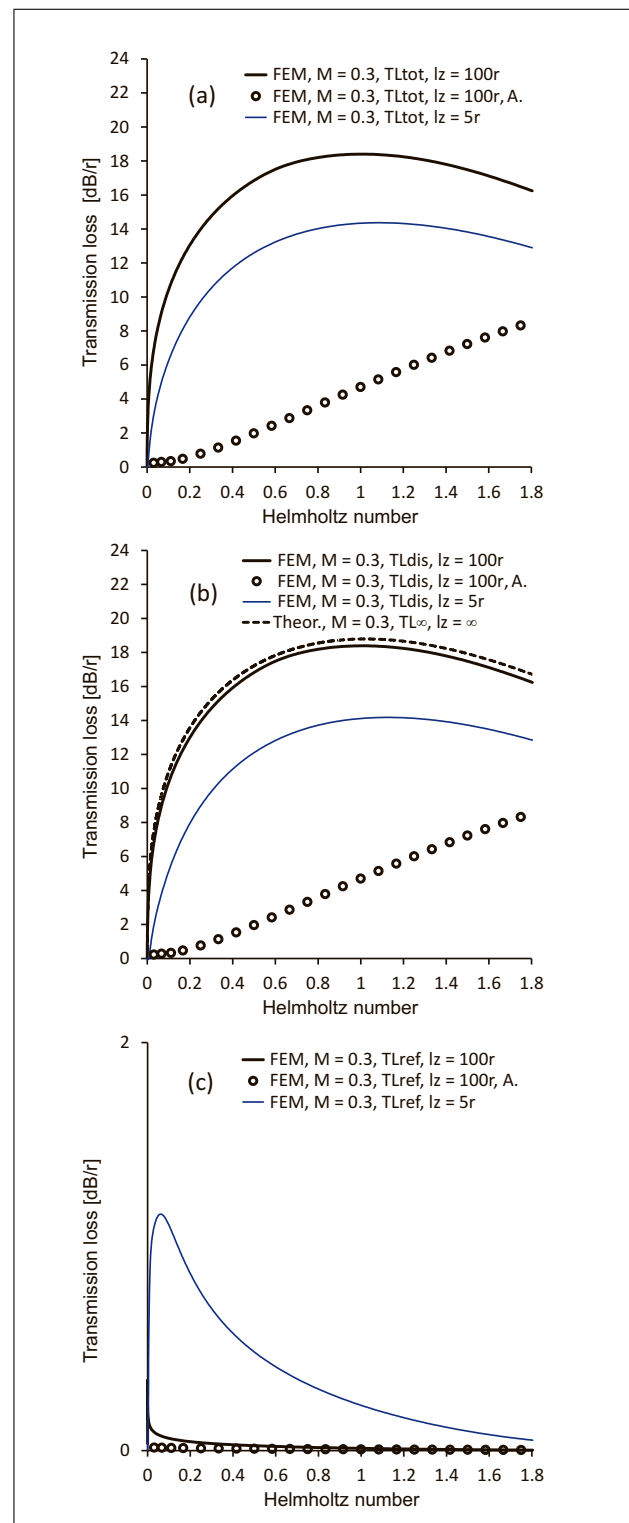


Figure 10. Dissipation versus reflection in the FEA results (Figure 7b) for flow ducts of different reduced lengths (section 2).

impedance leads to dominantly dissipative TL also in the case of finite length ducts.

The same trends, dominating high level of dissipation (Figure 11b) and negligible reflection (Figure 11c), are also observed in the case of the silencer prototype.

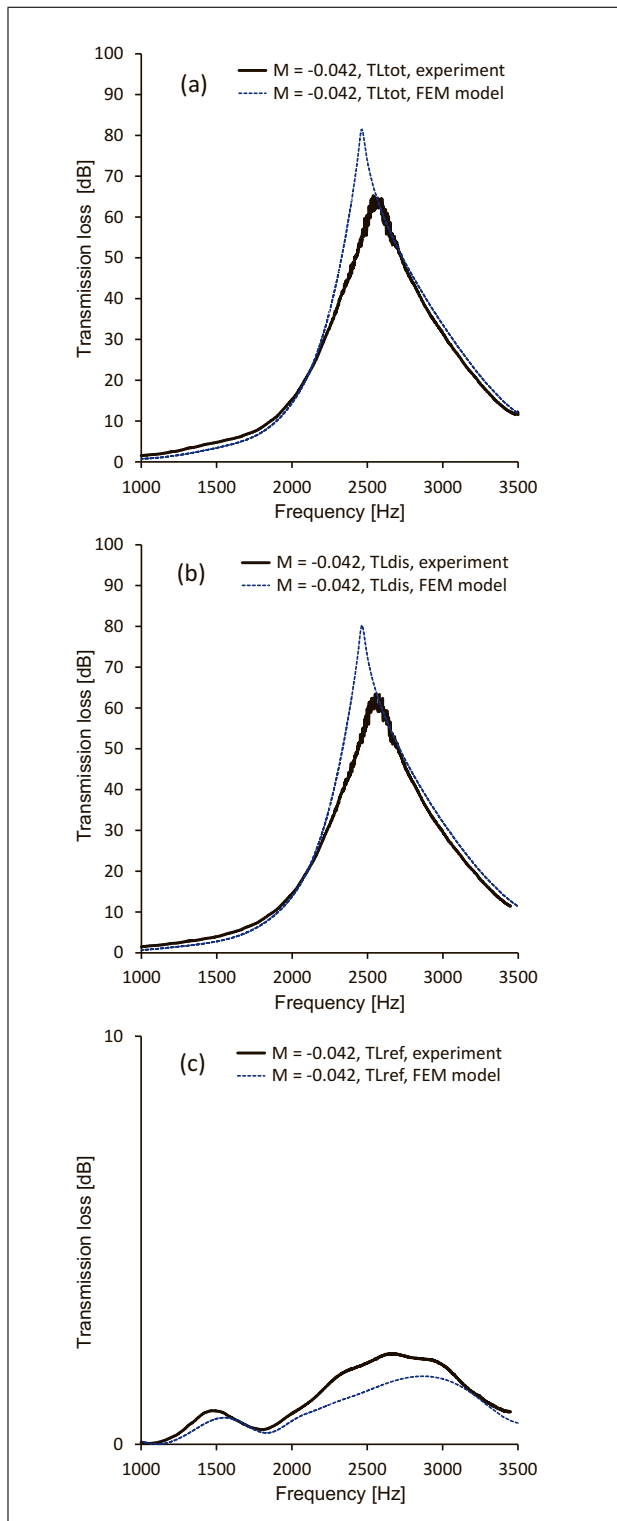


Figure 11. Dissipation versus reflection of the silencer prototype optimized using the Cremer impedance (section 3).

6. Summary and conclusions

In the present work, the Cremer optimal impedance model [1, 2, 14] is presented and analyzed from the perspective of non-fibrous silencer optimization. Theoretical results are that the original asymptotic model proposed by Tester [2],

leads to significant deviation from “exact” optimal radial wave number ($k_r r$) in presence of mean flow and frequencies well below cut-on (Figure 3). Based on these results, plots for the “exact” Cremer impedance (Figure 5) as well as the corresponding attenuation rates (Figure 4) are provided for a range of Mach-numbers. An interesting feature is that the optimal Cremer impedance requires a negative real part in the low frequency range, i.e., an active boundary. It also worth to note that the solution shows a significant difference in the low frequency behavior for the up- and down-stream cases. For the down-stream case the limiting radial wave-number is $(0 + 0i)$ implying no damping (see Figures 3 and 4). Further analysis of this result as well as the negative real part would be an interesting continuation of the present work. A numerical FEA model, (section 2) validated by experiments (section 3), is used to also study a finite length duct with a Cremer wall impedance. The results show that a significantly larger transmission loss can be achieved using the “exact” Cremer impedance rather than the asymptotic model (Figure 7).

Furthermore, a method to analyze dissipation versus reflection for ducted elements in the plane wave range (two-ports) is proposed in section 4. Applied to the optimal Cremer impedance, it shows that silencers based on this concept will have a transmission loss strongly dominated by dissipation (Figure 10 and 11), i.e. it represents a true dissipative type of silencer solution.

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Appendix

By using Equation (9), the Equation (3) reads

$$\frac{d}{d(k_r r)} \left[f_1^{-2} \frac{k_r r J'_m(k_r r)}{J_m(k_r r)} \right] = 0, \quad (A1)$$

then differentiating leads to

$$\begin{aligned} & -\frac{2}{f_1^3} \frac{df_1}{d(k_r r)} \frac{k_r r J_1(k_r r)}{J_0(k_r r)} \\ & + \frac{1}{f_1^2} \frac{[J_1(k_r r) + k_r r J'_1(k_r r)] J_0(k_r r) + k_r r J_1^2(k_r r)}{J_0^2(k_r r)} = 0, \end{aligned} \quad (A2)$$

and

$$\begin{aligned} & \frac{k_r r}{f_1^2 J_0(k_r r)} \left\{ -\frac{2}{f_1} \frac{df_1}{d(k_r r)} J_1(k_r r) \right. \\ & \left. + \left[\frac{J_1(k_r r)}{k_r r} + J'_1(k_r r) \right] + \frac{J_1^2(k_r r)}{J_0(k_r r)} \right\} = 0. \end{aligned} \quad (A3)$$

Finally, the expression inside the brackets takes the form of the Equation (10):

$$\frac{df_1}{d(k_r r)} \frac{2J_1(k_r r)}{f_1} - J_0(k_r r) - \frac{J_1^2(k_r r)}{J_0(k_r r)} = 0. \quad (\text{A4})$$

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